Differential Equations Workbook

First order equations

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CLASSIFYING DIFFERENTIAL EQUATIONS

■ 1. Identify the order and linearity of the differential equation.

$$5y''' + 4xy = x^2$$

■ 2. Say whether or not the linear equation is homogeneous.

$$y'' - x = 3y$$

■ 3. Identify the order and linearity of each differential equation.

$$x^3 + 3x \sin x = y'$$

$$x\frac{d^4y}{dx^4} - x \ln x = 0$$

$$y'' - 3y' + xy^2 = x$$

■ 4. Determine whether each linear equation is homogeneous or non-homogeneous.

$$y' - 3y = 0$$

$$(\sin x)y' = 0$$

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$$\ln x - 3y'' = y$$

■ 5. Identify the order and linearity of each differential equation, then say whether or not each linear equation is homogeneous.

$$5y' - y''' - \ln y = x$$

$$y'' = 4y^2$$

$$y' = xy$$

■ 6. Identify the order and linearity of the differential equation.

$$e^x y''' = e^y x$$

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LINEAR EQUATIONS

■ 1. Find the solution to the linear differential equation.

$$y' - y \sin x = 0$$

■ 2. Solve the differential equation.

$$y' \cos x + y \sin x = 1$$

■ 3. Find the solution to the linear differential equation.

$$xy' + y = e^x$$

■ 4. Solve the differential equation.

$$xy' = x^3 - 3x^3y$$

■ 5. Find the solution to the linear differential equation.

$$xy' - 3y = x^2 - x + 1$$

■ 6. Solve the differential equation.

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$$xy' - y = \sqrt{x}$$

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INITIAL VALUE PROBLEMS

■ 1. Solve the initial value problem if $y(0) = 0$.

$$5y' - 10xy = 25x$$

■ 2. Solve the initial value problem if $y(1) = -3$.

$$x\frac{dy}{dx} + 2y = 6x^2$$

■ 3. Solve the initial value problem if $y(0) = 1$.

$$y' - y = 2\sin(3x)$$

■ 4. A function $y(x)$ is a solution of the differential equation. Suppose that $y(1) = 1$ and $y(3) = 3$. Find the constant k and the solution $y(x)$.

$$y' - \frac{ky}{x} = 0$$

■ 5. Solve the initial value problem if $y(\ln 2) = 1$.

$$\frac{dy}{dx} - 6y = 2$$

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■ 6. A function $y(x)$ is a solution of $y' + x^k y = 0$. Suppose that $y(0) = 1$ and $y(1) = e^{-5}$. Find the constant k and the solution $y(x)$.

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SEPARABLE EQUATIONS

■ 1. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{3x^2}{y}$$

■ 2. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{x+2}{2y}$$

■ 3. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{4x^3 - 1}{x(y^2 - 1)}$$

■ 4. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{x^2\sqrt{x^3 - 3}}{y^2}$$

■ 5. Find the solution to the separable differential equation.

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$$y' + \sin(x - y) = \sin(x + y)$$

■ 6. Solve the initial value problem if $y(0) = -1$.

$$(xy^2 + x) + (x^2y - y)y' = 0$$

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SUBSTITUTIONS

■ 1. Use a substitution to solve the separable differential equation.

$$y' = \sin(x + y)$$

■ 2. Use a substitution to solve the separable differential equation.

$$y' = (x + y - 5)^2$$

■ 3. Use a substitution to solve the initial value problem, if $y(0) = 0$.

$$y' = e^{5y-x}$$

■ 4. Use a substitution to solve the separable differential equation.

$$y' = \frac{1}{6x - 3y}$$

■ 5. Use a substitution to solve the initial value problem, if $y(0) = 8$.

$$y' + (9x + y - 1)^2 = 0$$

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6. Use a substitution to solve the initial value problem, if $y(0) = 2$.

$$y' + 2x + 2y - 2xy = x^2 + y^2$$

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BERNOULLI EQUATIONS

■ 1. Find the solution to the Bernoulli differential equation.

$$(y^4 + x^4 y)\, dx - 3x^5\, dy = 0$$

■ 2. Find the solution to the Bernoulli differential equation.

$$xy' - 3y = xy^{\frac{5}{3}}$$

■ 3. Find the solution to the Bernoulli differential equation.

$$y' = y^2 \sin x + y \cot x$$

■ 4. Find the solution to the Bernoulli differential equation, if $y(1) = 0$.

$$y' - \frac{y}{x} + \sqrt{y} = 0$$

■ 5. Find the solution to the Bernoulli differential equation, if $y(1) = 1$.

$$y' + \frac{2}{x}y = \frac{y^2}{x^2}$$

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6. Find the solution to the Bernoulli differential equation, if $y(0) = 2$.

$$3y' + 2y = xy^5$$

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HOMOGENEOUS EQUATIONS

■ 1. Find the solution to the differential equation.

$$x^2yy' = x^3 + xy^2$$

■ 2. Use a substitution to find a solution to the homogeneous equation.

$$x(x+y)y' - y(2x+y) = 0$$

■ 3. Rewrite the homogeneous equation in terms of y/x, but don't solve the separable equation.

$$yy' - x = \sqrt{x^2 + y^2}$$

■ 4. Use a substitution to find a solution to the homogeneous equation.

$$xyy' = x^2 e^{-\frac{y^2}{x^2}} + y^2$$

■ 5. Find the solution to the differential equation, if $y(1) = 1$.

$$x^2y' = 3xy + 2y^2$$

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■ 6. Use a substitution to find a solution to the homogeneous equation, if $y(1) = 1$.

$$xy' + y \ln x - y = y \ln y$$

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EXACT EQUATIONS

■ 1. If the differential equation is exact, find its solution.

$$2xy + (x^2 - y^2)\frac{dy}{dx} = 0$$

■ 2. What is the solution to the exact differential equation?

$$(6y + xe^{-y})\ dy - e^{-y}\ dx = 0$$

■ 3. Find the value of a that make the equation an exact differential equation.

$$(5\sin y - x^3)\ dx + ax\cos y\ dy = 0$$

■ 4. If the differential equation is exact, find its solution.

$$(x\sin y - 2x)\ dx + \left(\frac{x^2}{2}\cos y - 2y\right)\ dy = 0$$

■ 5. Find the value of a that makes the equation an exact differential equation.

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$$(2\cos x + 4\ln y)\,dx + \left(\frac{a^2 x}{y} + e^y\right)\,dy = 0$$

■ 6. What is the solution to the exact differential equation?

$$(x\cos y + e^y)\,dy + (e^x + \sin y)\,dx = 0$$

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